1.AI/ML-Powered Phishing Website Detection System

What is the Project about?

The project aims to develop an **AI/ML-powered phishing website detection system** that protects users from malicious domains in **real-time**.

- It extracts features from the URL, domain, SSL certificate, and webpage content.
- Trains a Machine Learning model (e.g., Random Forest, XGBoost) to classify whether a site is **legitimate or phishing**.
- Integrates into a browser extension (Privasee) to **warn/block users before they access phishing pages**.
- Enhances **user trust** by providing **explainable reasons** (e.g., "Domain too new" or "Suspicious characters in URL").

Existing System and Drawbacks?

1. Existing System

- **Blacklist-based Detection**
  - Services like **PhishTank, Google Safe Browsing, OpenPhish** maintain a database of reported phishing domains.
  - Browsers/extensions compare the requested URL with the blacklist.
  - If present → block access.
- **Heuristic/Rule-based Detection**
  - Simple rules like: URL too long, contains `@` or IP address, no HTTPS, etc.
  - Works fast but can be easily bypassed.

2. Drawbacks

- **Blacklist Limitations**:
  - Can only block **already-reported phishing sites**.
  - Zero-day phishing sites (freshly created) won't be in the list yet → user still exposed.
- **Rule-based Limitations**:
  - Attackers evolve quickly → they avoid simple rules.
  - High false positives (legit sites may get flagged).
- **Lack of Explainability**:
  - Current systems just say *"Blocked for security reasons"*.
  - Users don't know **why** → leads to mistrust or ignorance.
- **Reactive, not proactive**:
  - They rely on detection **after** phishing is already successful elsewhere.

Our Proposed Solution:

A **Machine Learning-driven phishing detection system** that proactively identifies phishing sites by analyzing multiple layers of information (URL, domain, SSL, page content).

✅ **Proactive Defense**: Detects phishing even if not yet reported (zero-day protection).

✅ **Multi-feature Analysis**: Goes beyond just URL → also checks SSL, WHOIS, page structure.

✅ **Explainability**: Shows reasons (e.g., *"Domain registered 3 days ago + suspicious tokens in URL"*).

✅ **Integration with Privasee**: Unified privacy + phishing protection tool.

✅ **User Control**: Option to **block automatically** or **warn & let user decide**.

✅ **Continuous Learning**: Updates model as new phishing patterns emerge.

2. HealthBot – AI-Powered Healthcare Customer Support Chatbot

What is the Project about?

1. Objective:

- To provide instant, intelligent, and accurate responses to patient or healthcare user queries.
- To assist users in understanding healthcare services, medicine instructions, appointment scheduling, and general medical guidance (non-critical).

2. Core Functionality:

1. **User Query Handling:** Users type a question like:
    - "How do I take this medicine?"
    - "When is my next appointment?"
    - "What are the side effects of paracetamol?"
2. **Persistent Memory:** The bot stores all past queries and responses in a database (or vector DB).
    - If a user asks a repeated or similar question, the bot retrieves relevant past answers.
3. **Generative Responses:** The bot uses an LLM to generate new answers for queries that have not been asked before or to add extra context.
4. **Summarization:** Combines previous related answers into a concise summary to improve response quality.
5. **Additional Features:**
    - Analytics for common healthcare questions.
    - Admin dashboard for FAQs.
    - Optional adaptive tone based on user sentiment (friendly/serious).

3. Impact:

- Reduces wait times for healthcare support.
- Improves patient knowledge and engagement.
- Reduces workload on human support staff.

Existing Systems & Their Drawbacks

1. Examples of existing systems:

1. **Zendesk / Intercom (Healthcare Chatbots):**
    - Mostly FAQ-based, rule-based chatbots.
    - Limited memory — cannot "remember" previous conversations contextually.
2. **Symptom Checker Bots (Babylon Health, Ada Health):**
    - Focused on symptom diagnosis, not general healthcare support.
    - Limited customization for hospital-specific information.
3. **Hospital Websites / Helpdesk Bots:**
    - Often static or require manual updating.
    - Cannot provide personalized or context-aware answers.

2. Drawbacks:

- **No long-term memory:** Each interaction is independent; repeated questions require the same answer from scratch.
- **Generic responses:** Often fail to handle nuanced queries about medicine, appointments, or hospital policies.
- **Low adaptability:** Cannot improve over time using previous conversations without manual intervention.
- **No analytics:** Admins cannot easily track the most common queries or pain points.

Proposed Solution & How It Solves Drawbacks

Key Features:

1. **Persistent Memory:** Stores queries and bot responses in a database. Uses embeddings to retrieve contextually relevant past answers.
2. **Hybrid Retrieval + Generative Responses:**
   - For repeated or similar questions → retrieve old answers.
   - For new questions → generate answers using LLM.
   - Combine both intelligently for a coherent response.
3. **Healthcare-Specific Knowledge Base:**
   - Integrates hospital FAQ, medicine guides, appointment info, etc.
   - Ensures accurate and domain-specific responses.
4. **Analytics Dashboard:**
   - Track most frequent questions.
   - Identify knowledge gaps in FAQs.
   - Monitor user satisfaction (via feedback ratings).
5. **Optional Enhancements for Uniqueness:**
   - Adaptive tone or language (formal/friendly, multilingual).
   - Summarization of previous conversations to avoid overwhelming the user.
   - Multimodal support: users can upload prescription images to ask questions.

Why it's better:

- Provides **context-aware responses**.
- Reduces repetitive workload on staff.
- Continuously improves with each conversation.
- Niche focus ensures accurate, domain-specific support.

3. MediTrack – Smart Medicine Expiry Tracker

What is the Project about?

1. Objective:

- To help users (patients, pharmacists, households) **track medicine expiry dates** to ensure safety and avoid using expired medicines.
- To reduce medicine wastage and improve health safety awareness.

2. Core Functionality:

1. **Medicine Input:** Users can input medicines manually or scan them using **OCR** (Optical Character Recognition) to read expiry dates from the strip/box.
2. **Expiry Tracking:**
   - Stores medicine name, batch, and expiry date in a database.
   - Tracks remaining days until expiry.
3. **Alerts & Notifications:**
   - Sends reminders before a medicine expires (configurable alert period).
   - Can notify users when multiple medicines are nearing expiry.
4. **History & Analytics:**
   - Shows expired and near-expiry medicines.
   - Helps users manage stock, reduce waste, and avoid accidental consumption.

3. Impact:

- Ensures safe use of medicines.
- Helps households and pharmacies maintain inventory efficiently.
- Reduces financial loss due to wasted medicines.

Existing Systems & Their Drawbacks

1. Examples of existing solutions:

1. **Simple mobile reminders / calendar apps:**
   - Users have to manually enter expiry dates.
   - No automated scanning; prone to human error.
2. **Pharmacy inventory systems:**
   - Designed for commercial use; not for end-users.
   - Usually too complex for home use.
3. **Other medicine tracker apps:**
   - May not support partial strips or split packs.
   - Limited to text input, no OCR/image support.
   - Notifications may not be configurable.

2. Drawbacks:

- **Manual entry is tedious** and error-prone.
- **No intelligent detection** of cut strips or partial packs.
- **No visual aid / OCR support**, making it hard to track medicine in real life.
- **No analytics** for waste reduction or consumption patterns.

Proposed Solution & How It Solves Drawbacks

1. Key Features:

1. **OCR-Based Input:** Users can scan medicine strips or boxes. System automatically detects medicine name, batch, and expiry date.
2. **Partial Strip Handling:**

        1. Allows users to log leftover doses when strips are cut.
        2. Tracks expiry per batch or partial strip.
    - **Smart Alerts & Reminders:**
        1. Configurable notifications (e.g., 7 days before expiry).
        2. Visual alerts in-app for near-expiry medicines.
    - **Database & Analytics:**
        1. Tracks all medicines and expiry history.
        2. Can provide suggestions to avoid future wastage.
    - **Optional Enhancements:**
        1. Barcode scanning support.
        2. Integration with pharmacies for refill reminders.
        3. Voice-based medicine input for accessibility.

2. Why it's better:

- Reduces human error and manual entry.
- Tracks both full and partial packs accurately.
- Provides proactive alerts, helping users avoid health risks.
- Analytics and history provide **actionable insights** for users.